

Red

picoCTF > Forensics · + Tags

A png file was given in the question. Firstly, I check the metadata of the png file using `exiftool` command

```
(kali㉿kali) ~[~/ctf/red]
$ exiftool red.png
ExifTool Version Number      : 13.50
File Name                    : red.png
Directory                    : .
File Size                    : 796 bytes
File Modification Date/Time   : 2025:03:05 22:34:15-05:00
File Access Date/Time        : 2026:07:21 09:19:05-04:00
File Inode Change Date/Time   : 2026:07:21 09:19:05-04:00
File Permissions              : -rw-rw-r--
File Type                    : PNG
File Type Extension          : png
MIME Type                    : image/png
Image Width                  : 128
Image Height                 : 128
Bit Depth                   : 8
Color Type                   : RGB with Alpha
Compression                  : Deflate/Inflate
Filter                      : Adaptive
Interlace                   : Noninterlaced
Poem                        : Crimson heart, vibrant and bold,.Hearts flutter at your sight..Evenings glow softly red,.Cherries burst with sweet life..K
isses linger with your warmth..Love deep as merlot..Scarlet leaves falling softly,.Bold in every stroke.
Image Size                   : 128x128
Megapixels                   : 0.016
```

Only the poems seems fishy. I think I have to decode the poem or something. Lets type `Strings` to see the poem more clearly

```
(kali㉿kali) - [~/ctf/red]  
$ strings red.png  
IHDR  
tEXtPoem  
Crimson heart, vibrant and bold,  
Hearts flutter at your sight.  
Evenings glow softly red,  
Cherries burst with sweet life.  
Kisses linger with your warmth.  
Love deep as merlot.  
Scarlet leaves falling softly,  
Bold in every stroke.x  
IDATx  
IEND
```

Every First Letter of the poem leads to [CHECK LSB](#). Now, lets go to cyberchef steganography decoding tool to find the [LSB](#) needed.

[illegible]

The colour pattern hint was given in the question and there is only one cipher repeating which looks like a Base64 encryption.

Recipe

From Base64

Alphabet
A-Za-z0-9+/=

☒ Remove non-alphabet chars ☐ Strict mode

Input

cG1jb0NURntyM2RfMXNfdGgzX3VsdfNFZjByXzU0ZG4zNTVffQ==

Output

picoCTF{r3d_1s_th3_ult1m4t3_cur3_f0r_54dn355_}

```
flag: picoCTF{r3d_1s_th3_ult1m4t3_cur3_f0r_54dn355_}
```